* $S = \{2, 8, 5\}$, $|S| = 3$

For a finite set, its cardinality /size $\in \mathbb{Z}^+$

$\aleph_0 = |\mathbb{Z}^+| \rightarrow$ countable set

(aleph NULL)

→ First Hebrea Alphabet

Schroder - Bernstein Theorem:

$|A| = |B| \rightarrow |A| < |B|$, $|B| < |A|$

Prove that, $|[0,1)| = |[0,1]|$

$A = [0,1)$ $\qquad$ $B = [0,1]$

$f(x) = x$ $\qquad$ $g(x) = x/2$

Relation:

Let A & B be two sets then a binary relation R is a subset of $A \times B$.

$\boxed{R \subseteq A \times B}$ "ordered pair"

$$\left\{ \begin{array}{l} p+q<n \\ p+q=n \\ \frac{1}{n-1}, \frac{2}{n-2}, \frac{3}{n-3} \end{array} \right\}$$

Ex: 6: Set of real no.s is uncountable.

Proof by contradiction:

Assume that the set of $\mathbb{R}$ is countable, $[0,1)$ is a subset of $\mathbb{R}$.

Therefore, $[0,1)$ is also countable.

r_1	$0.$	d_1	d_2	d_3	d_4	d_5	d_6 .....
r_2	$0.$	d_1	d_2	d_3	d_4	d_5	d_6
r_3	$0.$	d_1	d_2	d_3	d_4	d_5	d_6
r_4					d_4		
r_5						d_5	
r_6							d_6

$d_i \in \{0, 1, 2, \ldots, 9\}$

To create a real number $r = d_1' d_2' d_3' \ldots$

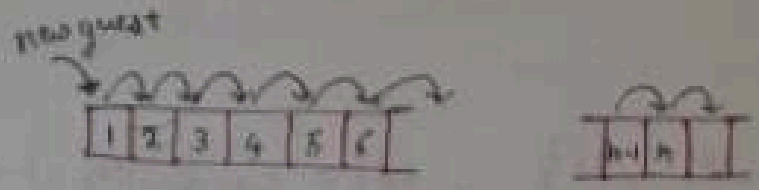
$d_i' = 4$ if $d_i \neq 4$

$= 8$ if $d_i = 4$

Claim: r is different from all r_i's $[0,1)$

d_i' is different from in the ith digit d_i.

($[0,1)$ is uncountable.)



Ex: 4: Set of integers is a countable set,

$$\{ \ldots\ldots -3 \ -2 \ -1 \ 0 \ 1 \ 2 \ 3 \ldots\ldots \}$$

$$(0 \quad 1)-(1 \quad 2)-(2 \quad 3)-3$$

$$f(n) = \frac{n}{2} \text{ if } n \text{ is even}$$

$$\frac{n-1}{2} \text{ if } n \text{ is odd}$$

1	2	3	4	5	6	7	8
↓	↓	↓	↓	↓			
0	1	-1	2	-2			

Ex: 5: Set of positive rational no.s is a countable set.

(Cantor's Diagonalization argument)

p/q ($q+$

gcd(p,

$\frac{1}{1}$	$\frac{2}{1}$	$\frac{3}{1}$	$\frac{4}{1}$	$\frac{5}{1}$
$\frac{1}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{4}{2}$	$\frac{5}{2}$
$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$	$\frac{5}{3}$
$1/4$	$2/4$	$3/4$	$4/4$	$5/4$
$1/5$	$2/5$	$3/5$	$4/5$	$5/5$

$p+q=2$ $\quad 1/1^{(1)}$

$p+q=3$ $\quad 2/1^{(2)}$ $1/2^{(3)}$

$p+q=4$ $\quad 1/3^{(4)}$ $3/1^{(5)}$

$p+q=5$ $\quad 3/2^{(6)}$ $2/3^{(7)}$ $1/4^{(8)}$ $4/1^{(9)}$

$p+q=6$ $\quad 1/5^{(10)}$ $5/1^{(11)}$

* Size of a Set:

$S = \{1, 5, 2, -2, 99\}$, $|S|$ = Cardinality of the set

For a finite set, we can establish one-to-one correspondance between a subset of positive integers $\{1, 2, 3, \ldots\}$ to S.

S is 5.

↳ **Countable Set:** It is either a finite set there is a one-to-one correspondance b/w the set of +'ve integers to the given set.

Ex: 1: Set of positive even integers is a countable set.

$$f(n) = 2n \qquad n \in \{1, 2, 3, \ldots\}$$

(one-to-one ≡ one-one / onto)

$$f(n) = f(m)$$

$$2n = 2m \Rightarrow n = m$$

$$g(n) = \frac{f(n)}{2}, \quad \text{where } f(n) \text{ is an even integer}$$

Ex: 2: Set of positive odd integers

$$f(n) = 2n - 1, \; n \in \{1, 2, 3, \ldots\}$$

Ex: 3: Proof that if a guest comes to Hilbert's Grand Hotel having infinite rooms, the guests can be accomodated.

David Hilbert

The other degrees comes from a vertex that must be on the maximal path.

$\overline{B \to A}$: (Claim)

G is a connected even graph

Proof: The claim using the method of induction on the no. of edges $|E|$ of G.

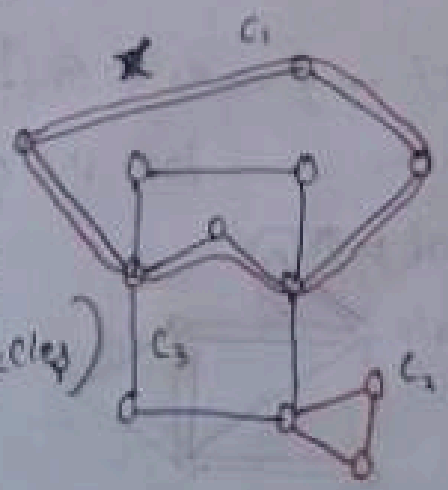
(0 is even)

Basis Step: $|E| = 0$, 0 ✓

* Induction Step Hypothesis $< |E|$

$B \to A$ is true for any connected Even subgraph of G whose no. of edges $< |E|$

* We have to prove that the claim is true for G having $|E|$ edges, we know that G is a connected even graph.



(Segregating as a set of cycles)

Euler Graph

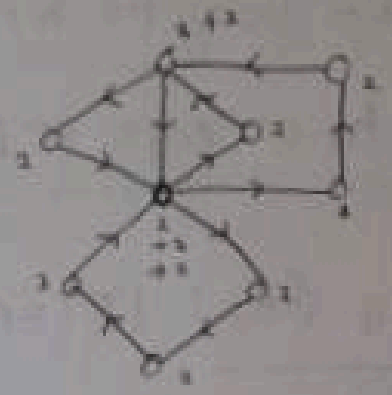
degree either same or $-2, -4$

(even connected graph $< |E|$)

[only if] $\underline{A \to B}$

G has an Euler tour.

→ closed trail is given that passes through each edge of G exactly once.

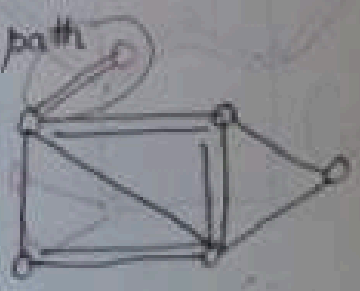


* For each vertex $v \in V - \{u\}$, I enter u through an edge xu and exit u using a different edge ~~* contri~~ uy, Thus passage through u contributes a degree 2 to u & for u we start and end our tour at v, which contributes $d(u) = 2$.

Proof by Extremality:

* <u>Lemma</u> : If every vertex of G has a degree at least 2, then the graph has a cycle.

1) Find a maximal path with end points u & v.



(maximal path)

For u/v, the edges in the path contributes to degrees.

Euler

* Theorem:

A tour is an Euler tour iff the graph has a non-trivial component, where degree of each vertex is even in component of

non-trivial: graph has atleast 1 edge.

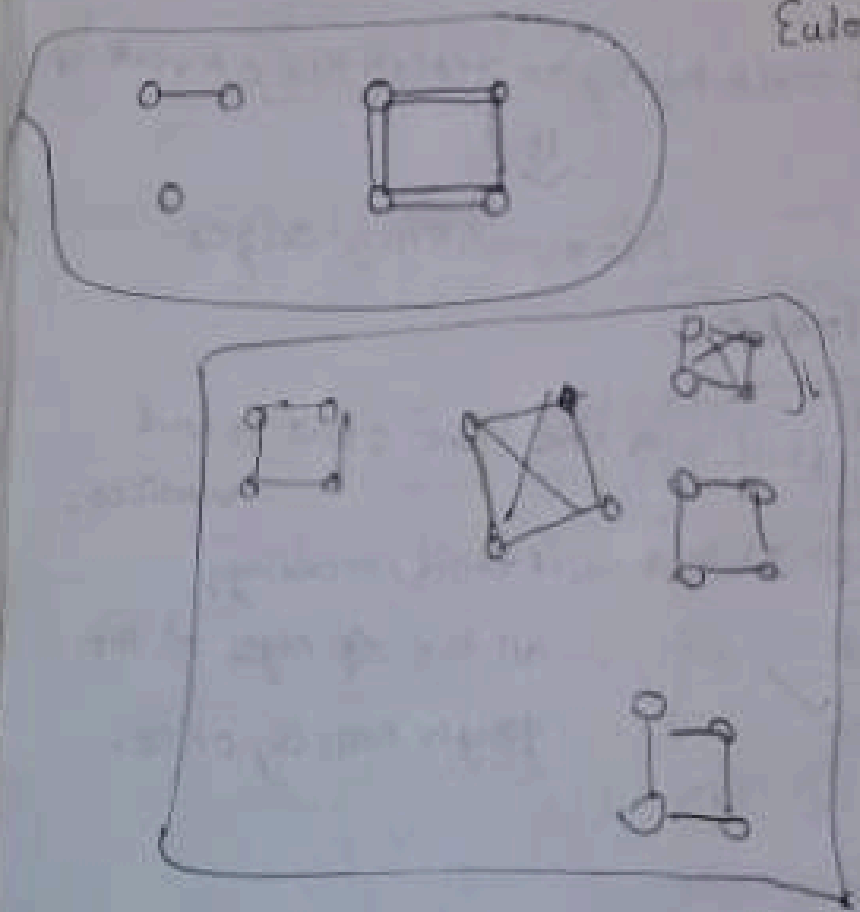
component ≡ maximally connected subgraph

trivial → o

non-trivial → o—o

→ (Connected undirected graph)

Euler Graph ⟷ Even connected graph



A ≡ G=(V, E) has a Euler tour

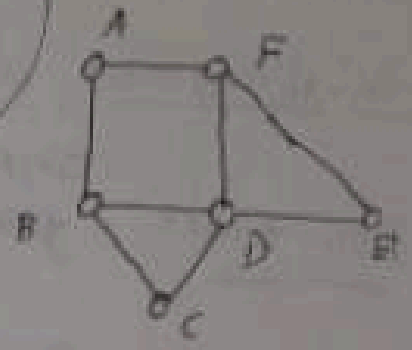
B ≡ G=(V,E) is a connected graph where vertices have even degree

A if B : B → A A ⟺ B

A only if B : A → B

$$\begin{bmatrix} M1 - W3 \\ M2 - W2 \\ M3 - W1 \\ M4 - W4 \end{bmatrix}$$

Euler: $(R = |E| - |V| + 2)$



Walk

Trail

Path

Walk: Visit of vertices and edges while respecting the neighbourhood properties of each vertex.

Trail: A walk with no edge repeated

Path: A walk having no repetition of vertices

$\Downarrow$

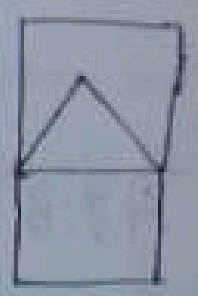
No repetition of edges

CLOSED TRAIL:

- A trail with the same start & end vertices.

Euler Tour: It is a closed trail containing, all the edges of the graph exactly once.

✗



✓

STABLE MARRIAGE / Matching

Kabhi Alvida Na Kehna

Shahrukh KHAN ⟷ Preeti Zinta

Abhishek Bachhan ⟷ Rani Mukherjee

Gale - Shapley : (Stable matching Algorithm)

M1 W3 W2 W1 W4

M2 W3 W2 W4 W1

M3 W4 W3 W1 W2

M4 W4 W3 W2 W1

Men propose Women - MAY BE

W1 M1 M2 M3 M4

W2 M1 M3 M2 M4

W3 M1 M3 M4 M2

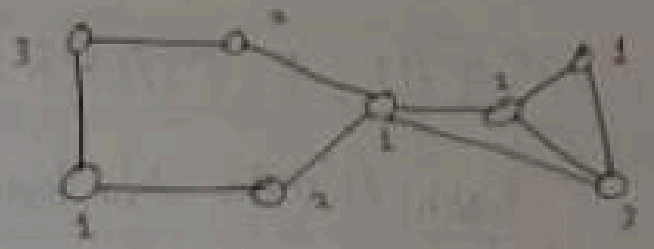
W4 M1 M4 M3 M2

M1 W3 maybe

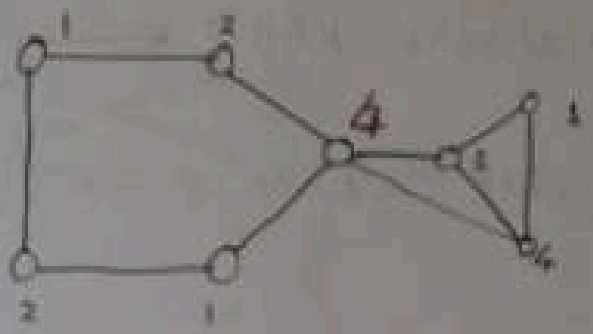
M2 W3 Rejected W2 maybe

M3 W4 ~~maybe~~ reject W3 reject

M4 W4 maybe



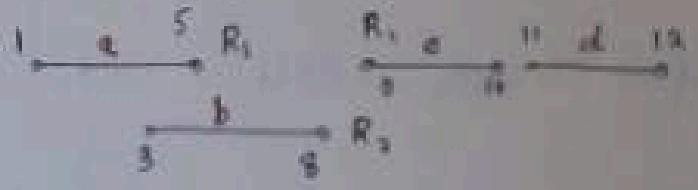
$\chi(G) = 3$



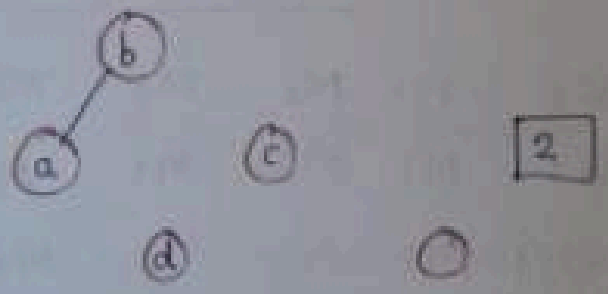
Register Allocation Problem :

(Interval graph) What is the minimum no. of registers needed to execute the program?

(lifetime = scope)



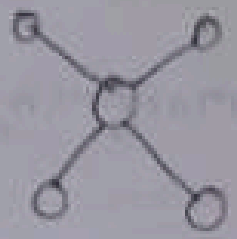
(interval coloring will give optimal answer)



$$\boxed{\Delta(G)+1 \geq \chi(G) \geq \max\left\{\omega(G), \frac{|V|}{\alpha(G)}\right\}}$$

$\Delta(G) \equiv$ Maximum degree of a vertex in a graph

$\delta(G) \equiv$ Minimum degree of a vertex in a graph



$\Delta(G)$ edges

even if it is not a complete graph then also $\Delta(G)$ colours required for neighbours & +1 for the present vertex.

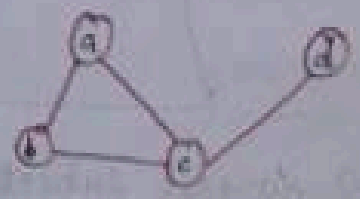
(For complete graph,

$\Delta(G) = n-1$

& $\chi(G) = \Delta(G)+1$)

Welsh - Powell Algorithm :

Q.// Where does WPA fail?



c	a	b	d
3	2	2	1

(1) Sort the vertices in non-increasing order of their degrees.

(2) For each vertex you assign the index that is not used to color its neighbors.

* Independant Set :

Given a graph , $G = (V, E)$, a subset of vertices

$U \subseteq V$ is called an independant set, if

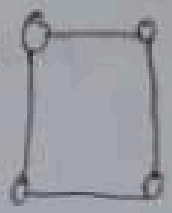
there is no edge among U.

no. of vertices $\Leftarrow$ $\alpha(G)$ = Independance No. of a graph.

longest subset with pairwise disjoint vertices

$\alpha(K_{m,n}) = \max(m, n)$, $\alpha(K_n) = 1$

$\alpha(\bar{K}_n) = n$



$n \equiv$ even

$n \equiv 2$

$\alpha(C_n) = \lfloor \frac{n}{2} \rfloor$

$\chi(G) \geq \omega(G)$

$n \equiv$ odd

$n \equiv 2$

$$\Rightarrow \boxed{\chi(G) \geq \frac{|V|}{\alpha(G)}}$$

$\alpha(G) \Rightarrow$ longest subset of disjoint vertices

[lower bounds of $\chi(G)$] $\boxed{\chi(G) = 3, \alpha(G) = 4}$

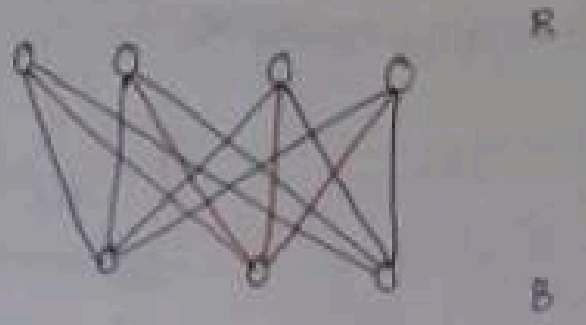
R – 4

G – 4

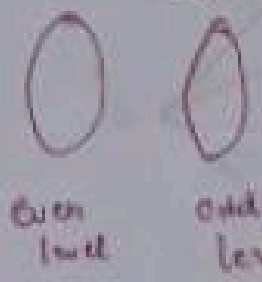
B – 4

$$\chi(G) \times \alpha(G) \geq |V|$$

$$\chi(K_{m,n}) = 2$$



(Every tree is bipartite)



For a tree, $\chi = 2$ //

$\chi(T_n) = 2$

(if not bipartite, at least 3 colors)

(cycle)

$\chi(C_n) = \{$

bipartite:

→ no cycle

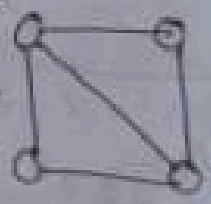
→ cycle (even)

$\chi(P_n) = 2$

(path)

* Clique number (ω):

No. of vertices in the largest subgraph of a graph, which is a clique.



$$\chi(G) \geq \omega(G)$$

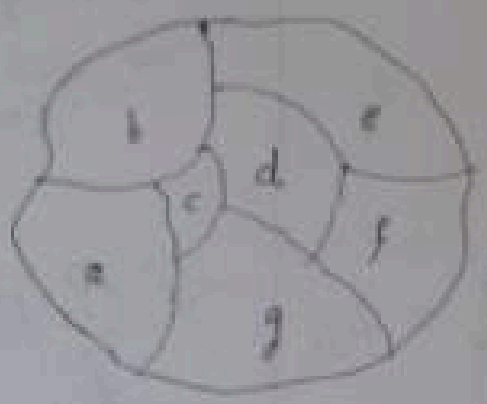
$\omega($

$\chi($

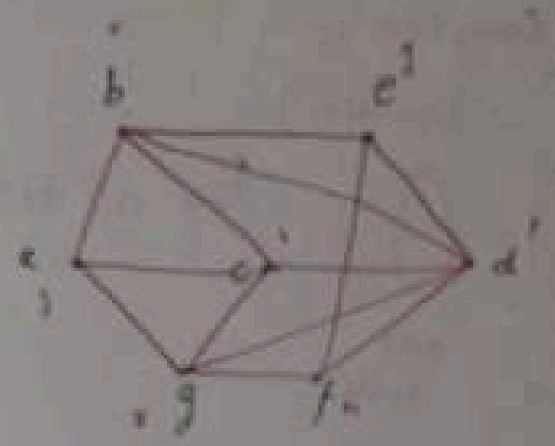
Colouring of Graph:

- → Vertex Coloring ✓
- → Edge Coloring ✗

Vertex Coloring: Assignment of of colours to the vertices of graph such that adjacent vertices receives different colors.



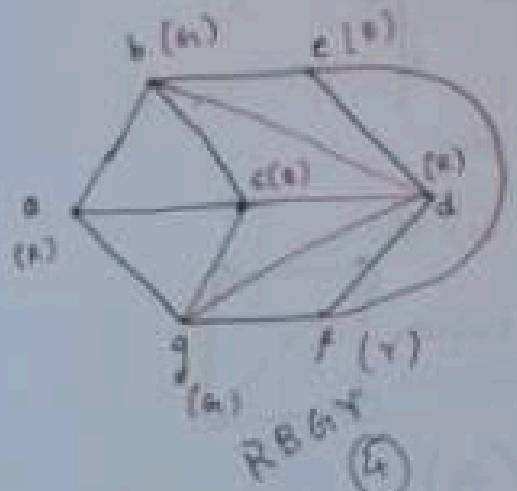
⇒



χ (chi) → Chromatic Number of a graph

⇓

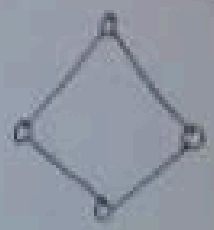
Minimum no. of colors needed for proper colouring

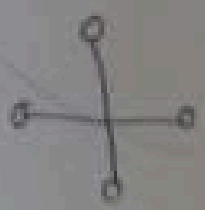


(planar graph ≡ 4 colour enough)

$(\chi(K_n) = n)$ $(\chi(\overline{K_n}) = 1)$

⊛ Compliment Graph:



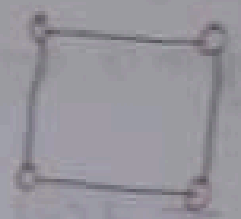


$G \cup G^c = K_n$

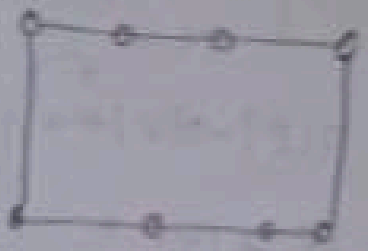
K (reduction)T

Kuratowski's Theorem:

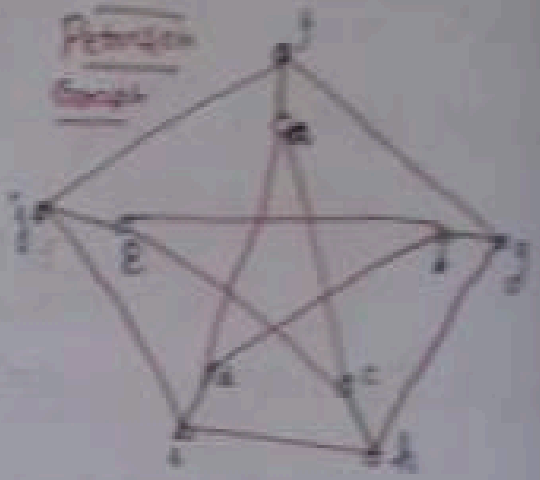
A graph is not planar iff the graph has a sub graph which is homeomorphic to either $K_{3,3}$ or K_5.

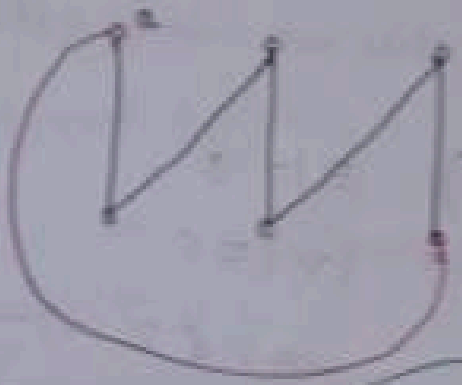


* ~~as~~ may ~~con~~ test as you wish



Petersen Graph





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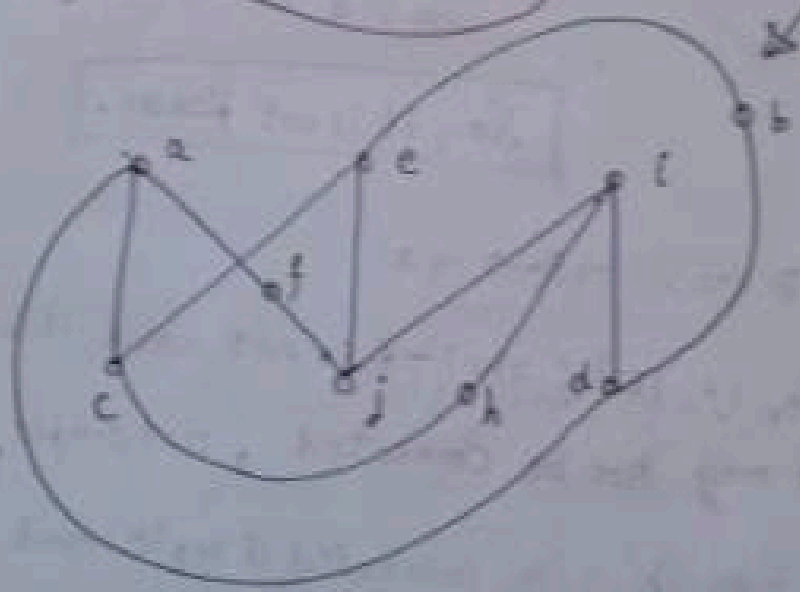

(4-colour / 5-colour) : Theorem